

SQL Learning Journal #14- MULTIPLE JOIN

Introduction

Today I learned how to combine information from more than two tables using multiple ****INNER JOIN**** clauses. I also learned how to use additional matching conditions with the ****AND**** operator in the ****ON**** clause.

SQL Commands Learned Today

1. INNER JOIN ON and INNER JOIN ON function

Table_1:

name	year	age
John	1999	26
Barbro	2001	24
Kristin	1996	29

Table_2:

name	year	city
John	1999	Madrid
Kenneth	2007	Istanbul
Kristin	1996	Köln

Table_3:

city	population
Madrid	3400k
Köln	1100k
Istanbul	16k

SELECT t1.name, t1.year, t2.city, population

FROM Table_1 AS t1

INNER JOIN Table_2 AS t2

ON Table_1.name=Table_2.name

INNER JOIN Table 3 AS t3

ON Table_2.city=Table_3.city

The same query can also be written using the ****USING**** clause.

SELECT t1.name, t1.year, t2.city, population AS city_population

FROM Table_1 AS t1

INNER JOIN Table_2 AS t2

USING (name)

INNER JOIN Table 3 AS t3

USING (city)

RESULT TABLE:

name	year	city	city_population
John	1999	Madrid	3400k
Kristin	1996	Köln	1100k

2. AND function with INNER JOIN ON function

Additional matching conditions can be added by using the ****AND**** operator inside the ****ON**** clause.

Table_1:

name	year	age
John	1999	26
Barbro	2001	24
Kristin	1996	29

Table_2:

name	year	city
John	1999	Madrid
Kenneth	2007	Istanbul
Kristin	2005	Köln

```

SELECT *
FROM Table_1 AS t1
INNER JOIN Table_2 AS t2
ON Table_1.name=Table_2.name
    AND Table_1.year=Table_2.year

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Result_table:

name	year	city
John	1999	Madrid

What I Learned Today

Today I learned how to join more than two tables by using multiple ****INNER JOIN**** clauses. I also practiced adding extra matching conditions with the ****AND**** operator inside the ****ON**** clause. These techniques make SQL queries more powerful when working with relational databases.

In real-world databases, information is often stored in multiple related tables. Multiple INNER JOINS allow us to combine customer, order, product, or location data into a single query result.